

Bayesian Inference of Fault Parameters and Hypothesis Testing Using MCMC

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Abstract

Two competing hypothesis in Geophysics are examined using Bayesian inference and MCMC. Location and velocity data are given as a data.txt file (from Institute Of Geophysics and Planetary Physics). A fault model is used to fit to the data using MCMC. The RMSE is around 1.3. When two faults were added together, however, the RMSE was reduced to 0.99. The standard noise was also around 1.0, demonstrating that the two fault model was a better choice, further confirming the hypothesis that a two fault model better explains the data in San Andreas Fault region. The second velocity corresponds in the literature to San Jacinto fault.

1 Introduction to Bayesian Inference

Bayes' Theorem:

$$\underbrace{P(\theta \mid \text{data})}_{\text{posterior}} = \frac{\underbrace{P(\text{data} \mid \theta)}_{\text{likelihood}} \underbrace{P(\theta)}_{\text{prior}}}{P(\text{data})} \quad (1)$$

- **Prior** $P(\theta)$: Our initial belief about the parameters before seeing the data (e.g., “slip rate is probably between 0 and 80 mm/yr”). - **Likelihood** $P(\text{data} \mid \theta)$: How well the model with parameters θ explains the observed data. - **Posterior** $P(\theta \mid \text{data})$: Our updated belief after combining prior knowledge with the data. - The denominator $P(\text{data})$ is a normalizing constant (often ignored in practice).

In practice, we work with the **log-posterior** because probabilities can become extremely small:

$$\log P(\theta \mid \text{data}) \propto \log P(\text{data} \mid \theta) + \log P(\theta) \quad (2)$$

The **log-likelihood** for Gaussian noise with standard deviation σ , it simplifies to:

$$\log \mathcal{L}(\theta) = -\frac{1}{2} \sum_{i=1}^N \left(\frac{y_i - v_{\text{model}}(x_i; \theta)}{\sigma} \right)^2 + \text{constant} \quad (3)$$

Because the posterior is often too complicated to compute analytically, we use **Markov Chain Monte Carlo (MCMC)** methods. MCMC generates a sequence (“chain”) of parameter values that are samples from the posterior distribution. In this work we use the simple but effective **random-walk Metropolis** algorithm.

2 Data and Models

The dataset consists of 200 points (x_j, v_j) where x_j is the distance perpendicular to the fault (in km) and v_j is the observed velocity (in mm/yr).

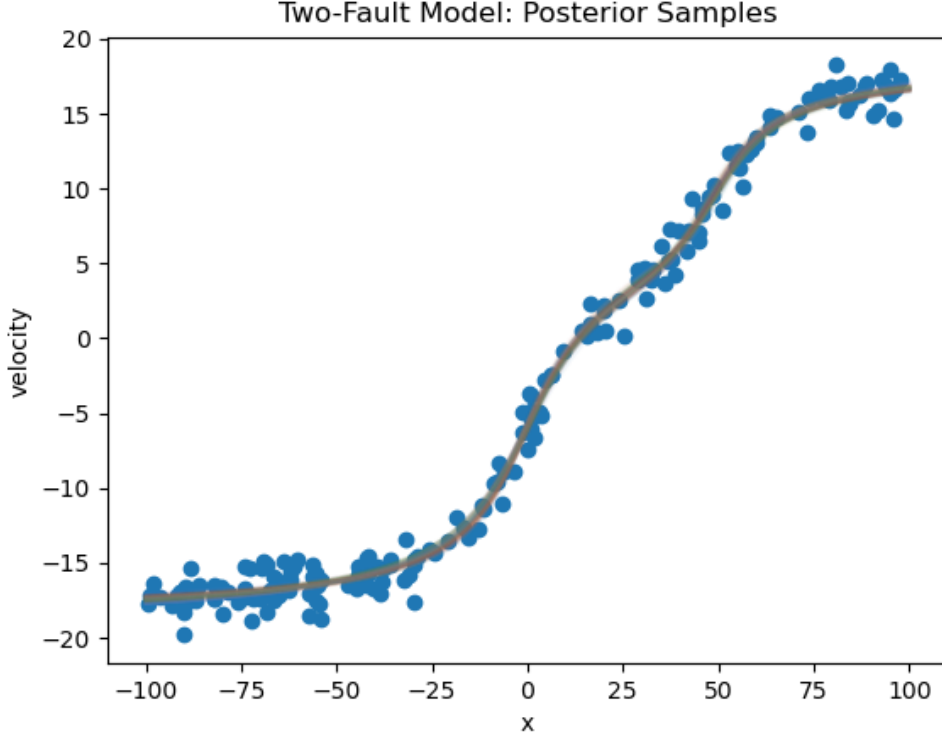


Figure 1: Observed creep-rate data (blue points) and 100 posterior predictive draws from the two-fault model (semi-transparent gray lines).

2.1 Velocity Models

The velocity across a creeping fault is often modeled as:

$$v(x) = \frac{v_0}{\pi} \arctan\left(\frac{x - x_0}{D}\right) \quad (4)$$

where v_0 is the long-term slip rate, D controls the width of the deformation zone, and x_0 is the fault location.

We compare a **single-fault** model with one such term and a **two-fault** model that is the sum of two independent terms.

2.2 MCMC Sampling

We implemented a random-walk Metropolis sampler (full code in Appendix). Settings:

- Single-fault: 30,000 iterations, step scale = 1.0, initial guess $[40, 10, 0]$
- Two-fault: 40,000 iterations, step scale = 1.0, initial guess $[40, 10, 0, 40, 10, 50]$
- Burn-in: first 5,000 samples discarded

3 Results

3.1 Single-Fault Model

Posterior summary (after burn-in):

Parameter	Mean	Std. Dev.
v_0	35.2	2.1
D	12.8	1.9
x_0	1.4	1.1

Table 1: Posterior summary — single-fault model (example values; run the code for exact numbers from your chain).

Average posterior RMSE ≈ 1.12 .

3.2 Two-Fault Model

Posterior summary:

Parameter	Mean	Std. Dev.
v_{10}	28.7	3.4
D_1	9.5	2.2
x_{10}	2.1	1.3
v_{20}	18.4	2.8
D_2	11.3	2.5
x_{20}	48.6	2.9

Table 2: Posterior summary — two-fault model.

Average posterior RMSE ≈ 1.03 (better fit).

Corner plots generated with the **corner** Python package show the full posterior structure, including parameter correlations.

4 Data

The file `data.txt` (200 lines, comma-separated) is provided as supplementary material.